

Storage Services

Study Guide

AMARJIT SEAL
AI&DS, PIET
Parul University

UNIT 1: Types of Cloud Storage

Cloud storage is the foundation of modern computing. AWS offers three primary types of storage, each optimized for different use-cases and workloads.

1.1 Object Storage

Object storage is a storage architecture that manages data as discrete units called objects. Each object consists of three parts:

- Data – The actual content (file, image, video, document, etc.)
- Metadata – Descriptive information about the object (size, type, creation time, custom tags)
- Unique Identifier – A globally unique key/URL to access the object

Key Characteristics of Object Storage:

- Flat structure (no folder hierarchy — folders are simulated via key prefixes)
- Accessed via HTTP/HTTPS using REST APIs (PUT, GET, DELETE operations)
- Highly scalable — can store petabytes to exabytes of data
- Designed for unstructured data such as images, videos, backups, and log files
- High durability — data is replicated across multiple locations
- Eventual consistency model — changes may take time to propagate

Real-World Use Cases:

- Backup and disaster recovery
- Static website hosting
- Data lakes and big data analytics
- Content delivery (images, videos)
- Archival and compliance storage

 **Note:** Object storage cannot be used as a boot disk for an operating system. It is NOT mountable like a file system.

1.2 Block Storage

Block storage manages data in fixed-size blocks (like a traditional hard disk drive). The operating system sees block storage as a raw disk and can format it with any file system.

Key Characteristics of Block Storage:

- Data is divided into uniformly sized blocks (e.g., 4KB, 512B)
- Each block has its own address — no metadata stored with data
- Attached to a server like a virtual hard disk drive (HDD/SSD)
- Very low latency — ideal for I/O-intensive applications
- Requires OS-level file system formatting (ext4, NTFS, XFS, etc.)
- Supports read/write operations at the block level

Real-World Use Cases:

- Database storage (MySQL, PostgreSQL, Oracle)
- Boot volumes for virtual machines (EC2 instances)
- Enterprise applications requiring high IOPS
- Transactional workloads

 **Note:** EBS (Elastic Block Store) is AWS's block storage service. A single EBS volume can only be attached to ONE EC2 instance at a time (except Multi-Attach for io1/io2).

1.3 File Storage

File storage organizes data in a hierarchical tree structure with folders and files — similar to a traditional network file system (NFS or SMB). Multiple clients can access the same file system simultaneously.

Key Characteristics of File Storage:

- Hierarchical directory/folder structure
- Accessed via standard file protocols: NFS (Network File System) or SMB (Server Message Block)
- Supports concurrent access by multiple clients/instances
- POSIX-compliant — supports permissions, locking, and file attributes
- Files are organized in directories with metadata

Real-World Use Cases:

- Shared content repositories
- Home directories for multiple users
- Media workflows with shared assets
- Development environments with shared code
- Content Management Systems (CMS)

 **Note:** Amazon EFS (Elastic File System) is AWS's managed file storage service. It supports only Linux-based instances via NFS protocol.

1.4 Comparison: Object vs Block vs File Storage

Feature	Amazon S3	Amazon EBS	Amazon EFS
Type	Object Storage	Block Storage	File Storage (NFS)
Access Method	HTTP/HTTPS API (PUT/GET)	Attached to ONE EC2 instance	Mounted to MULTIPLE EC2s
Performance	Moderate (web latency)	High (low latency)	Good (network-based)
Scaling	Unlimited / automatic	Manual resize required	Automatic (elastic)
OS Support	Any (via API/SDK)	Linux & Windows	Linux only (NFS)
Durability	99.999999999% (11 nines)	99.999% (with snapshots)	Highly durable (replicated)
Availability Zones	Multi-region globally	Single AZ (same as EC2)	Multi-AZ (Regional)
Pricing Model	Per GB + requests	Per GB provisioned	Per GB stored (pay-as-you-go)
Best Use Cases	Backups, websites, data lakes	Databases, boot volumes	Shared storage, CMS, dev envs

 **Tip:** Memory Trick — S3 is 'Object' (think S3 = Scalable, Simple, Spread everywhere). EBS is 'Block' (think B for Block, Boot). EFS is 'File' (think F for File, Folder, Flexible).

UNIT 2: Amazon S3 (Simple Storage Service)

Amazon S3 is AWS's core object storage service. It provides industry-leading scalability, data availability, security, and performance. S3 is one of the most widely used services on AWS.

2.1 Core Features of Amazon S3

- Unlimited storage capacity — stores any amount of data
- Objects can be up to 5 TB in size; uploaded in parts (Multipart Upload for >100MB)
- Data is stored in Buckets — globally unique containers
- Accessed via HTTP/HTTPS REST APIs or AWS CLI/SDK
- 99.999999999% (11 nines) durability — data replicated across minimum 3 Availability Zones
- 99.99% availability for Standard storage class
- Supports versioning, lifecycle policies, event notifications

2.2 S3 Buckets

Creating a Bucket:

1. Log into the AWS Management Console
2. Navigate to S3 service
3. Click 'Create Bucket'
4. Provide a globally unique bucket name (e.g., myapp-storage-2024)
5. Select the AWS Region closest to your users
6. Configure public access settings (blocked by default — recommended)
7. Enable versioning if required
8. Click 'Create Bucket'

S3 Bucket Naming Rules:

- Must be globally unique across all AWS accounts
- 3–63 characters long
- Lowercase letters, numbers, and hyphens only
- Must start and end with a letter or number
- Cannot be formatted as an IP address (e.g., 192.168.5.4)

2.3 S3 Key Features

Versioning:

Versioning preserves, retrieves, and restores every version of every object stored in a bucket. Once enabled, all objects get a version ID. Deleted objects are marked with a Delete Marker (not permanently deleted).

S3 Lifecycle Policies:

Lifecycle rules automatically transition objects between storage classes or delete them after a specified time period. Example: Move to Standard-IA after 30 days, to Glacier after 90 days, delete after 365 days.

S3 Replication:

- Cross-Region Replication (CRR): Automatically replicates objects to a bucket in a different region for compliance or disaster recovery.
- Same-Region Replication (SRR): Replicates objects within the same region for log aggregation or data sovereignty.

S3 Event Notifications:

S3 can trigger events when objects are created, deleted, or restored. Events can invoke AWS Lambda, send to SQS, or publish to SNS. Common use: trigger a Lambda function to process an image when it is uploaded to S3.

S3 Transfer Acceleration:

Uses AWS CloudFront edge locations to speed up uploads to S3. Files are uploaded to the nearest edge location and then routed to S3 over AWS's optimized network backbone.

S3 Encryption:

- SSE-S3 (Server-Side with S3 Keys): AWS manages keys automatically. Default encryption option.
- SSE-KMS (Server-Side with KMS Keys): Uses AWS Key Management Service. Provides audit trail and fine-grained access control.
- SSE-C (Server-Side with Customer Keys): Customer provides and manages encryption keys.
- Client-Side Encryption: Data is encrypted before being uploaded to S3.

 **Note:** SSE-KMS is preferred when you need to control access to encryption keys while AWS manages key rotation.

2.4 S3 Security

- **Bucket Policies:** JSON-based resource policies attached directly to the S3 bucket. Can allow/deny access based on IP, user, or AWS account.
- **Access Control Lists (ACLs):** Legacy mechanism to set permissions on buckets and objects.
- **IAM Policies:** Control which AWS users/roles can access S3 resources.
- **Block Public Access:** Four settings to prevent public exposure of buckets and objects (recommended to keep all enabled for most use cases).
- **Pre-signed URLs:** Temporary URLs that grant time-limited access to specific objects without requiring AWS credentials.

UNIT 3: S3 Storage Classes

Amazon S3 offers multiple storage classes designed for different use cases and access patterns. Choosing the right storage class optimizes cost while meeting performance and availability requirements.

3.1 Overview of All S3 Storage Classes

Storage Class	Use Case	Availability	Min Duration	Retrieval
S3 Standard	Frequently accessed data	99.99%	None	Instant
S3 Intelligent-Tiering	Unknown/changing access	99.9%	None	Instant
S3 Standard-IA	Infrequently accessed	99.9%	30 days	Instant (per-GB fee)
S3 One Zone-IA	Infrequent, non-critical	99.5% (1 AZ)	30 days	Instant
S3 Glacier Instant	Archives, ms retrieval	99.9%	90 days	Milliseconds
S3 Glacier Flexible	Archives, mins–hrs retrieval	99.99%	90 days	1 min–12 hrs
S3 Glacier Deep Archive	Long-term archival	99.99%	180 days	12–48 hrs
S3 Express One Zone	Single-digit ms latency	99.95% (1 AZ)	None	Instant

3.2 Detailed Storage Class Descriptions

1. S3 Standard (Default)

- Designed for frequently accessed data
- 99.999999999% durability, 99.99% availability
- Replicated across minimum 3 Availability Zones
- Lowest latency and highest throughput performance
- No minimum storage duration — pay only for what you use
- Best for: Active workloads, web applications, data analytics

2. S3 Intelligent-Tiering

- Automatically moves objects between access tiers based on usage
- No retrieval fees and no minimum storage duration
- Four access tiers: Frequent, Infrequent, Archive Instant, Deep Archive
- Small monthly monitoring fee per object
- Best for: Data with unknown or changing access patterns

3. S3 Standard-IA (Infrequent Access)

- For data accessed less frequently but needs rapid access when required
- Lower storage cost than S3 Standard, but has per-GB retrieval fee
- Minimum storage duration of 30 days
- Replicated across multiple AZs (same durability as Standard)
- Best for: Backups, disaster recovery files, older data

4. S3 One Zone-IA

- Same as Standard-IA but data stored in a SINGLE Availability Zone
- 20% cheaper than Standard-IA (due to reduced redundancy)
- Risk: Data lost if the AZ is destroyed
- Best for: Secondary backups, easily reproducible data

5. S3 Glacier Instant Retrieval

- For archive data that needs millisecond retrieval
- Same performance as Standard but much lower storage cost
- Minimum storage duration: 90 days
- Per-GB retrieval fee applies
- Best for: Medical images, news media assets, genomics data

6. S3 Glacier Flexible Retrieval (formerly Glacier)

- For archive data with retrieval times from 1 minute to 12 hours
- Three retrieval options: Expedited (1–5 min), Standard (3–5 hrs), Bulk (5–12 hrs)
- Minimum storage duration: 90 days
- Best for: Backup data, compliance archives

7. S3 Glacier Deep Archive

- Lowest cost storage class in AWS — designed for long-term retention
- Retrieval time: 12 hours (Standard) or 48 hours (Bulk)
- Minimum storage duration: 180 days

- Replicated across 3+ AZs — same durability as S3 Standard
- Best for: Regulatory compliance, digital preservation (7–10 year retention)

8. S3 Express One Zone

- Newest storage class — delivers single-digit millisecond latency
- Stored in a single AZ using S3 Directory Buckets
- 10x faster than S3 Standard for compute-intensive workloads
- Best for: ML training, real-time analytics, high-frequency data access

3.3 S3 Lifecycle Policies — Transition Rules

Lifecycle policies define transition timelines between storage classes:

- S3 Standard → Standard-IA: minimum 30 days after creation
- S3 Standard-IA → Glacier: minimum 30 days in Standard-IA
- Any class → Glacier Deep Archive: anytime after minimum durations
- Objects in One Zone-IA cannot transition to Glacier Instant Retrieval

 **Note:** Intelligent-Tiering is the BEST choice when access patterns are unpredictable. The monitoring fee (~\$0.0025 per 1,000 objects) is worth the cost savings.

UNIT 4: Amazon EFS (Elastic File System)

Amazon EFS is a fully managed, scalable, and elastic NFS file system for use with AWS cloud services and on-premises resources. It is designed to be used with Amazon EC2 Linux instances.

4.1 Key Features of Amazon EFS

- Fully managed — no servers to provision or manage
- Elastic storage — automatically grows and shrinks as files are added/removed
- Supports the NFS v4 protocol (Network File System version 4 and 4.1)
- Can be mounted to THOUSANDS of EC2 instances simultaneously
- Regional service — accessible across multiple Availability Zones
- POSIX-compliant file system — supports standard file permissions and locking
- Supports both Linux-based EC2 instances and on-premises servers (via Direct Connect or VPN)

4.2 EFS Performance Modes

1. General Purpose Mode (Default):

- Optimized for latency-sensitive use cases
- Ideal for web serving, content management, home directories
- Recommended for most workloads

2. Max I/O Mode:

- Scales to higher aggregate throughput and IOPS
- Slightly higher latency per operation
- Ideal for big data analytics, media processing, genomics

4.3 EFS Throughput Modes

- Bursting Throughput: Throughput scales with file system size. Accumulates burst credits when idle.
- Provisioned Throughput: Specify throughput independent of storage amount. Pay for the provisioned throughput.
- Elastic Throughput (recommended): Automatically scales throughput up or down based on workloads.

4.4 EFS Storage Classes

- EFS Standard: For frequently accessed files — highest availability and durability
- EFS Standard-IA (Infrequent Access): Lower cost for files not accessed every day
- EFS One Zone: Single AZ storage — lower cost, suitable for dev/test environments
- EFS One Zone-IA: Lowest cost option — single AZ + infrequent access

4.5 Mounting EFS to EC2

9. Create an EFS file system in the AWS Console
10. Configure mount targets in each Availability Zone
11. Ensure Security Groups allow NFS traffic on port 2049
12. Install the amazon-efs-utils package on the EC2 instance
13. Mount command: `sudo mount -t efs fs-XXXXXXXX:/mnt/efs`
14. For auto-mount on reboot: add entry to `/etc/fstab`

 **Note:** EFS requires port 2049 (NFS) to be open in the Security Group of both the EC2 instance and the EFS mount target.

UNIT 5: Amazon EBS (Elastic Block Store)

Amazon EBS provides persistent block-level storage volumes for use with Amazon EC2 instances. EBS volumes behave like raw, unformatted block devices and are attached to EC2 instances in the same Availability Zone.

5.1 Key Features of Amazon EBS

- Persistent storage — data persists independently of the EC2 instance lifecycle
- Attached to a SINGLE EC2 instance in the same Availability Zone
- Supports snapshots for backup (stored in Amazon S3)
- Can be encrypted using AWS KMS
- Can be dynamically resized (increase size only, not decrease)
- Supports Multi-Attach for io1/io2 volumes (multiple EC2 instances in same AZ)
- Designed for low-latency, high-performance I/O operations

5.2 EBS Volume Types

SSD-Backed Volumes (for IOPS-intensive workloads):

- gp2 (General Purpose SSD): Baseline 3 IOPS/GB, burst up to 3,000 IOPS. Max 16,000 IOPS.
- gp3 (General Purpose SSD v3): Latest generation. 3,000 IOPS baseline (can provision up to 16,000 IOPS independently). 20% cheaper than gp2. RECOMMENDED default choice.
- io1 (Provisioned IOPS SSD): High-performance. Up to 64,000 IOPS. For I/O-intensive databases.
- io2 Block Express: Highest performance SSD. Up to 256,000 IOPS. Sub-millisecond latency. For SAP HANA, Oracle.

HDD-Backed Volumes (for throughput-intensive workloads):

- st1 (Throughput Optimized HDD): Up to 500 MB/s throughput. For big data, data warehouses, log processing. Cannot be used as boot volume.
- sc1 (Cold HDD): Lowest cost. Up to 250 MB/s. For infrequently accessed data. Cannot be used as boot volume.

 **Note:** gp3 is now the recommended default EBS volume type as it provides better performance at lower cost compared to gp2.

5.3 EBS Snapshots

Snapshots are point-in-time backups of EBS volumes stored durably in Amazon S3.

- Incremental backups: Only changed blocks since the last snapshot are saved
- Can be used to create new EBS volumes (in same or different AZ/region)
- Can be copied to different AWS regions for DR purposes
- Can be shared with other AWS accounts
- Managed via AWS Backup or directly in the EBS console

5.4 Creating an Amazon EBS Volume — Step-by-Step

Method 1: Using AWS Management Console

15. Log in to AWS Console → Navigate to EC2 → Left panel: Elastic Block Store → Volumes
16. Click 'Create Volume'
17. Select Volume Type: Choose gp3 (recommended for general purpose)
18. Set Size: Specify in GiB (e.g., 20 GiB for a general workload)
19. Select IOPS: For gp3, default is 3,000 IOPS (can be increased up to 16,000)
20. Select Throughput: Default 125 MB/s for gp3
21. Select Availability Zone: MUST be same AZ as your EC2 instance
22. Encryption: Enable encryption (recommended) — choose KMS key
23. Add Tags: e.g., Name: MyDataVolume, Environment: Production
24. Click 'Create Volume'

Method 2: Attach the Volume to an EC2 Instance

25. In EBS Volumes list, select the newly created volume
26. Click Actions → Attach Volume
27. Select the target EC2 instance from the dropdown
28. Choose device name (e.g., /dev/xvdf or /dev/sdf)
29. Click 'Attach Volume'

Method 3: Format and Mount (Linux EC2)

- Check volume: `lsblk`
- Format: `sudo mkfs -t ext4 /dev/xvdf`
- Create mount point: `sudo mkdir /data`
- Mount: `sudo mount /dev/xvdf /data`
- Auto-mount on reboot: Add to `/etc/fstab`

 **Note:** If you stop an EC2 instance, the EBS root volume persists. If you terminate the instance, the root EBS volume is deleted by default (unless you disable 'Delete on Termination').

UNIT 6: Hosting a Static Website Using Amazon S3

Amazon S3 can host static websites — websites that contain fixed content such as HTML, CSS, JavaScript, and image files. This is a highly cost-effective and scalable solution that does NOT require any web server.

6.1 What is a Static Website?

A static website serves pre-built files to users. It does NOT involve server-side processing.

Technologies used:

- HTML (structure), CSS (styling), JavaScript (client-side interactivity)
- Images, videos, audio files
- JSON data files

S3 static websites CANNOT run server-side code (PHP, Python, Node.js). For dynamic content, integrate with AWS Lambda + API Gateway.

6.2 Step-by-Step: Host a Static Website on S3

Step 1: Create an S3 Bucket

30. AWS Console → S3 → Create Bucket
31. Bucket name should match your domain name (e.g., www.mywebsite.com) for custom domain support
32. Choose Region close to your audience
33. UNCHECK 'Block all public access' (required for public website)
34. Click 'Create Bucket'

Step 2: Enable Static Website Hosting

35. Open the bucket → Click 'Properties' tab
36. Scroll to 'Static website hosting' section → Click 'Edit'
37. Select 'Enable'
38. Index document: enter 'index.html' (main page)
39. Error document: enter 'error.html' (404 page)
40. Click 'Save changes'
41. Note the Endpoint URL provided (e.g., http://www.mywebsite.com.s3-website-us-east-1.amazonaws.com)

Step 3: Configure Bucket Policy for Public Access

42. Go to bucket → 'Permissions' tab → 'Bucket Policy'

43. Add the following JSON policy:

```
{ "Version": "2012-10-17", "Statement": [{ "Sid": "PublicReadGetObject",  
  "Effect": "Allow", "Principal": "*", "Action": "s3:GetObject",  
  "Resource": "arn:aws:s3:::your-bucket-name/*" }] }
```

Step 4: Upload Website Files

44. Go to bucket → 'Objects' tab → Click 'Upload'

45. Upload index.html, error.html, CSS, JS, and image files

46. Set permissions: Grant public read access during upload

47. Click 'Upload'

Step 5: Test the Website

48. Go to Properties → Static Website Hosting → Copy the endpoint URL

49. Open in browser — your website is live!

6.3 Optional Enhancements

- Custom Domain: Use Amazon Route 53 to point a custom domain to the S3 website endpoint
- HTTPS: Use Amazon CloudFront in front of S3 to enable HTTPS (S3 static websites only support HTTP)
- CDN: CloudFront caches and delivers content from edge locations globally for faster performance
- Logging: Enable S3 server access logging to track website requests

 **Note:** Amazon S3 static website hosting only supports HTTP. To serve content over HTTPS, you MUST use Amazon CloudFront as a CDN in front of S3.

6.4 Supported File Types for S3 Static Website Hosting

- HTML files and web pages (.html, .htm)
- Client-side scripts (.js)
- Stylesheets (.css)
- Image files (.jpg, .png, .gif, .svg, .webp)
- Video and sound files (.mp4, .mp3)
- PDF documents (.pdf)
- NOT supported: Server-side scripts (PHP, Python), dynamic content generation

UNIT 7: Important Question Bank

Compiled from AWS Certification Exams, University Exam Patterns, and Cloud Computing Syllabi. These questions cover all key topics.

Section A: Short Answer Questions (2–5 Marks)

Q1. What is the difference between Object Storage, Block Storage, and File Storage?

Answer: Object storage manages data as objects (S3), block storage as raw blocks (EBS), and file storage as hierarchical files (EFS).

Explanation: Object storage is accessed via HTTP APIs, block storage is attached as a disk to VMs, and file storage is mounted as a shared network drive. Each serves different performance and access pattern requirements.

Q2. What is Amazon S3? List any four key features.

Answer: Amazon S3 (Simple Storage Service) is a fully managed object storage service.

Explanation: Key features: (1) Unlimited storage capacity, (2) 11 nines durability (99.999999999%), (3) Multiple storage classes for cost optimization, (4) Versioning and lifecycle policies. Additional features include encryption, replication, and event notifications.

Q3. List and briefly explain all S3 storage classes.

Answer: S3 Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant, Glacier Flexible, Glacier Deep Archive, S3 Express One Zone.

Explanation: Each class is optimized for different access patterns and cost requirements. Standard is for frequent access; Glacier classes are for archival. Intelligent-Tiering automatically moves objects between tiers.

Q4. What is Amazon EBS? What are the different EBS volume types?

Answer: Amazon EBS is a persistent block storage service for EC2. Volume types: gp2, gp3 (SSD general purpose), io1, io2 (provisioned IOPS SSD), st1 (throughput HDD), sc1 (cold HDD).

Explanation: gp3 is the newest and recommended general-purpose type. io2 provides the highest performance for mission-critical databases. st1 and sc1 are for sequential, high-throughput workloads.

Q5. What is Amazon EFS? How does it differ from Amazon EBS?

Answer: EFS is a managed NFS file system that can be shared across multiple EC2 instances; EBS is block storage attached to a single EC2 instance.

Explanation: EFS scales automatically, supports concurrent access from thousands of instances, and is regional (multi-AZ). EBS is tied to a single AZ, must be manually resized, and attaches to one instance at a time.

Q6. How do you host a static website on Amazon S3?

Answer: Create S3 bucket, enable Static Website Hosting in Properties, configure bucket policy for public read access, upload HTML/CSS/JS files, and access via the S3 website endpoint.

Explanation: S3 static hosting supports HTML, CSS, JavaScript, and media files. It does NOT support server-side code. For HTTPS, use CloudFront in front of S3.

Section B: Multiple Choice Questions (MCQs)

Q7. Which statement correctly describes AWS storage services? (a) S3 is file, EBS is block, EFS is object (b) S3 is object, EBS is block, EFS is file (c) S3 is block, EBS is object, EFS is file (d) S3 is file, EBS is object, EFS is block

Answer: (b) S3 is object storage, EBS is block storage, and EFS is file storage.

Explanation: This is a fundamental concept. S3 stores objects via HTTP API, EBS is a virtual hard disk (block), and EFS is a shared network file system.

Q8. A fleet of EC2 instances across multiple AZs needs to access shared file-based data. Which service is best? (a) Amazon S3 (b) Amazon EFS (c) Amazon EBS (d) EC2 Instance Store

Answer: (b) Amazon EFS — it supports concurrent access from multiple EC2 instances across Availability Zones.

Explanation: EBS can only attach to one instance at a time. S3 is object storage accessed via API. EFS is the only choice for a shared, mountable file system across multiple EC2 instances.

Q9. A company has archive data that is rarely accessed but needs to be retrieved within minutes. Which S3 class is most cost-effective? (a) S3 Standard (b) S3 Glacier Deep Archive (c) S3 Glacier Instant Retrieval (d) S3 Standard-IA

Answer: (c) S3 Glacier Instant Retrieval — millisecond retrieval for archive data.

Explanation: Glacier Deep Archive takes 12–48 hours. Standard and Standard-IA are more expensive. Glacier Instant Retrieval is designed for archives needing immediate access.

Q10. What types of content does Amazon S3 support for static web hosting? (Select THREE) (a) HTML and image files (b) Client-side scripts (c) Server-side PHP scripts (d) Dynamic HTML (e) Video and sound files

Answer: (a), (b), (e) — HTML/images, client-side scripts, and media files.

Explanation: S3 static hosting does NOT support server-side scripts (PHP, Python, Node.js) or dynamic HTML generation. It only serves pre-built static content.

Q11. Which EBS volume type provides the highest IOPS performance? (a) gp3 (b) io1 (c) io2 Block Express (d) st1

Answer: (c) io2 Block Express — up to 256,000 IOPS with sub-millisecond latency.

Explanation: io2 Block Express is the highest-performance EBS volume type, designed for critical databases like SAP HANA and Oracle. gp3 supports up to 16,000 IOPS.

Q12. An EC2 instance is terminated. What happens to the data on the root EBS volume? (a) Moved to S3 (b) Deleted (by default) (c) Kept as a snapshot (d) Attached to another instance

Answer: (b) Deleted by default — root EBS volume is deleted when the EC2 instance is terminated unless 'Delete on Termination' is disabled.

Explanation: This is a critical behavior to remember. Data EBS volumes (non-root) are NOT deleted by default when the instance terminates.

Section C: Long Answer / Descriptive Questions (10–15 Marks)

Q13. Explain Amazon S3 storage classes in detail. How would you design a lifecycle policy to minimize storage costs for log files?

Answer: Discuss all 8 storage classes with use cases, then design: Standard (0–30 days) → Standard-IA (30–90 days) → Glacier Flexible (90–365 days) → Delete (after 1 year).

Explanation: Focus on transition rules, minimum storage durations, retrieval fees, and the trade-off between cost and access speed. For logs: active logs in Standard, older logs in Standard-IA, archives in Glacier.

Q14. Describe the step-by-step process to create an Amazon EBS volume and attach it to an EC2 instance. Include best practices for sizing, type selection, and encryption.

Answer: Create volume (type, size, AZ matching EC2, encryption) → Attach to EC2 → Format (mkfs) → Mount → Update /etc/fstab. Best practices: use gp3, enable encryption, take regular snapshots.

Explanation: Key points: AZ must match, gp3 preferred over gp2, always encrypt sensitive data, incremental snapshots for backup.

Q15. Compare and contrast Amazon S3, Amazon EBS, and Amazon EFS. For each service, provide two real-world scenarios where it would be the best choice.

Answer: S3: object storage, HTTP access, unlimited scale — use for website hosting, backup/archival. EBS: block storage, single EC2 — use for database volumes, boot volumes. EFS: file storage, multi-EC2 NFS — use for shared code repositories, media assets.

Explanation: Remember the keyword triggers: S3=object/static website, EBS=boot/database/single instance, EFS=shared/multiple instances/NFS.

Q16. Explain how to host a static website on Amazon S3. What are the limitations of S3 static website hosting and how can these be overcome using other AWS services?

Answer: Steps: Create bucket → Enable static hosting → Set bucket policy → Upload files → Access endpoint. Limitations: HTTP only (use CloudFront for HTTPS), no server-side code (use Lambda+API Gateway), no custom error pages for dynamic apps.

Explanation: Key integrations: CloudFront for HTTPS and CDN caching, Route 53 for custom domains, Lambda for dynamic functionality.

Section D: Scenario-Based Questions

Q17. A startup wants to store user profile photos, serve a static landing page, and run a high-traffic MySQL database. Which AWS storage service would you recommend for each use case and why?

Answer: Photos → Amazon S3 (object storage, HTTP accessible, unlimited scale). Landing page → S3 static hosting. MySQL → Amazon EBS (gp3 or io1 for low-latency block storage attached to EC2).

Explanation: Match each use case to the appropriate storage type. S3 for unstructured/web content, EBS for database/OS volumes.

Q18. A company stores 50 TB of logs. Logs are accessed daily for the first 30 days, then rarely. They must be retained for 7 years for compliance. Design the most cost-effective storage strategy.

Answer: 0–30 days: S3 Standard. 30–90 days: S3 Standard-IA. 90–365 days: S3 Glacier Flexible. After 1 year: S3 Glacier Deep Archive. Delete after 7 years. Use a Lifecycle Policy to automate transitions.

Explanation: Use lifecycle policies to automate transitions. Deep Archive is cheapest (~\$0.00099/GB/month). Cost drops significantly with each tier transition.

Q19. Multiple EC2 instances across 3 AZs need to share the same dataset. Users report seeing different data depending on which instance serves the request. What is the cause and the solution?

Answer: Cause: Each EC2 instance has its own EBS volume with separate copies of data. Solution: Replace EBS with Amazon EFS so all EC2 instances mount the same shared file system.

Explanation: This is a classic EBS vs EFS question. EBS cannot be shared across AZs. EFS is regional and supports concurrent access from instances across multiple AZs.

Q20. What S3 encryption should you use when: (a) your company policy requires audit logs of who decrypted what data, and (b) you want AWS to manage everything automatically?

Answer: (a) SSE-KMS — provides CloudTrail audit logs for every encryption/decryption event. (b) SSE-S3 — AWS fully manages keys with no additional configuration.

Explanation: SSE-KMS enables fine-grained access control and complete audit trail via CloudTrail. SSE-S3 is the simplest option with no user management required.

UNIT 8: Quick Revision Cheatsheet

8.1 Key Numbers to Remember

- S3 durability: 99.999999999% (11 nines)
- S3 Standard availability: 99.99%
- Max S3 object size: 5 TB (use Multipart Upload for >100 MB)
- EFS port: 2049 (NFS protocol)
- gp3 baseline IOPS: 3,000 | Max: 16,000
- io2 Block Express max IOPS: 256,000
- S3 Standard-IA minimum storage: 30 days
- Glacier Instant/Flexible minimum storage: 90 days
- Glacier Deep Archive minimum storage: 180 days
- S3 bucket name: 3–63 characters, globally unique

8.2 Keyword Decision Guide for Exams

If you see these keywords...	Choose this service	Why
object storage, static website, data lake, HTTP access, backup	Amazon S3	S3 is the object storage service
boot volume, database storage, single instance, low-latency, IOPS, block-level	Amazon EBS	EBS is block storage for single EC2
shared file system, multiple instances, NFS, POSIX, Linux, auto-scaling storage	Amazon EFS	EFS is a shared network file system
archive, rarely accessed, long-term retention, compliance	S3 Glacier (choose based on retrieval time)	Glacier classes = lowest cost archival
unknown access pattern, auto-optimization, no retrieval fees	S3 Intelligent-Tiering	Auto-moves between tiers based on access

8.3 Common Exam Traps

- EBS vs EFS: EBS = one instance. EFS = many instances. If question says 'multiple EC2 instances need shared access' → always EFS.
- S3 for boot disk: WRONG — S3 cannot be used as an OS boot volume. Only EBS can be a boot/root volume.

- S3 HTTPS: S3 static websites only support HTTP. For HTTPS, add CloudFront.
- EBS Multi-Attach: io1/io2 volumes can attach to multiple instances in the SAME AZ. This is not true for gp2/gp3.
- Glacier retrieval time: Glacier Instant = ms. Glacier Flexible = 1 min to 12 hrs. Deep Archive = 12–48 hrs.
- Delete on Termination: Root EBS volume is deleted when EC2 terminates (by default). Data EBS volumes are NOT.
- EFS supports only Linux. For Windows shared storage, use Amazon FSx for Windows File Server.

8.4 AWS Storage Services Summary

Property	S3	EBS	EFS
Storage Type	Object	Block	File (NFS)
Access Protocol	HTTP REST API	OS-level disk	NFS port 2049
Instances	Any (via internet)	1 EC2 (same AZ)	Many EC2s (multi-AZ)
Scalability	Unlimited auto	Manual resize	Elastic auto
OS	Any	Linux & Windows	Linux only
Pricing	Per GB + requests	Per GB provisioned	Per GB stored

All the Best for Your Exam!  | Prepared with AWS Official Documentation & University